

ASSIGNMENT #06

Course: CHE 301: CHEMICAL ENGINEERING THERMODYNAMICS
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Abstract

This report presents a complete and detailed thermodynamic analysis for CHE 301 Assignment #06. The report focuses on two main topics: gas-mixture saturation with water using the Peng–Robinson equation of state, and fugacity calculations for a liquid organic mixture under both ideal-mixture and EOS-based assumptions.

Problem A concerns a natural gas mixture containing methane, ethane, propane, iso-butane, and normal-butane. The first task is to determine the maximum amount of water that the gas can hold at 65 °F and 900 psia, reported in the industry units of lb/MMSCF of dry gas. The second task is to determine the dew-point and bubble-point temperatures of the saturated gas stream at the same pressure. The physical basis of the calculation is that, at saturation, the fugacity of water in the vapor phase must equal the fugacity of pure liquid water at the same temperature and pressure. The Peng–Robinson EOS was used for both the hydrocarbon mixture and water, and the composition of the saturated stream was determined by adjusting the water content until the fugacity residual vanished.

Problem B concerns a tertiary mixture of methanol, ethanol, and normal butanol at 25 °C and 20 bar with overall molar composition [0.2, 0.3, 0.5]. The first objective is to verify that the mixture is liquid by calculating the bubble-point and dew-point temperatures under two models: first an ideal-mixture model with the Poynting correction retained, and second the Peng–Robinson EOS model. The second objective is to calculate the fugacities of the pure components and the components inside the mixture for both models. This report develops the thermodynamic reasoning, the Excel workflow, the CHE Calculator function sequence, the numerical tables, and the final interpretation of the results.

Goals

- Develop the thermodynamic background needed for Assignment #06.
- For Problem A, convert the given natural-gas composition from weight fractions to mole fractions.
- Determine the maximum amount of water the gas can hold at 65 °F and 900 psia, and report the answer in lb/MMSCF.
- Calculate the dew-point and bubble-point temperatures of the saturated natural gas stream at 900 psia.
- Show how the water-content calculation is performed using a fugacity-equilibrium condition based on the Peng–Robinson equation of state.
- For Problem B, verify the liquid-phase behavior of the methanol/ethanol/*n*-butanol mixture by calculating dew-point and bubble-point temperatures.
- Perform the phase-boundary calculations for Problem B first with the ideal-mixture model and then with the Peng–Robinson EOS.
- Calculate the fugacities of the pure components and the component fugacities in the mixture for both the ideal and EOS-based models.
- Present the Excel strategy, CHE Calculator functions, equations, tables, screenshots, and final answers in a clean and organized report.

1. Problem Statement

1.1 Problem A: Gas Mixture Saturation with Water

A natural gas stream contains the following mass composition:

60.847 wt% methane,
14.256 wt% ethane,
8.363 wt% propane,
8.267 wt% iso-butane,
8.267 wt% normal-butane.

The gas is at

$$T = 65\text{ }^{\circ}\text{F}, \quad P = 900\text{ psia.}$$

Assume that:

- the natural gas mixture with and without water follows the Peng–Robinson equation of state,
- all binary interaction parameters are taken as zero,
- the thermal properties of pure water are also calculated with the Peng–Robinson EOS.

Calculate:

1. The maximum amount of water the gas can hold, in lb/MMSCF of dry gas.
2. The dew-point and bubble-point temperatures, in degrees Fahrenheit, of the saturated natural-gas stream at 900 psia.

1.2 Problem B: Fugacity Calculations

Consider a tertiary mixture of methanol(1), ethanol(2), and normal butanol(3) at

$$T = 25\text{ }^{\circ}\text{C}, \quad P = 20\text{ bar}$$

with overall molar composition

$$\mathbf{z} = [0.2, 0.3, 0.5].$$

Calculate:

1. The bubble-point and dew-point temperatures assuming first that the mixture is ideal and then assuming that the mixture follows the Peng–Robinson EOS. For the ideal-mixture case, do not neglect the Poynting factor.
2. The fugacities, in bar, of the pure components and of the components in the mixture assuming first that the mixture is ideal and then assuming that the mixture follows the Peng–Robinson EOS.

2. Background Knowledge and Governing Theory

2.1 Why This Assignment Builds on Multiple Chapters

This assignment combines ideas from several chapters of CHE 301. The overall logic is:

- **Chapter 1:** write the equations in residual form and solve them numerically.
- **Chapter 4:** use equations of state for real fluids and calculate fugacity coefficients.
- **Chapter 5:** use ideal-mixture vapor–liquid equilibrium concepts, dew-point and bubble-point calculations, and the Poynting correction.
- **Chapter 6:** extend VLE and fugacity calculations to non-ideal mixtures described with the Peng–Robinson EOS.

2.2 Physical Meaning of Saturation in Problem A

Problem A is a gas-saturation problem. The natural gas is initially dry, and then water is conceptually added until the gas reaches its saturation limit. The gas is just saturated when it contains the maximum amount of water that can remain in the vapor phase without forming a separate liquid-water phase.

At that limiting condition, the equilibrium requirement is

$$f_{H_2O}^V = f_{H_2O}^L$$

where $f_{H_2O}^V$ is the fugacity of water in the vapor-phase natural gas mixture and $f_{H_2O}^L$ is the fugacity of pure liquid water at the same temperature and pressure.

2.3 Why Fugacity Is Used Instead of Pressure

Because the system is treated with the Peng–Robinson equation of state, it is not handled as ideal. In that case, fugacity replaces simple pressure as the thermodynamically correct equilibrium quantity. For a component i in a real mixture,

$$f_i = \hat{\phi}_i x_i P \quad \text{or} \quad f_i = \hat{\phi}_i y_i P$$

depending on the phase composition used.

2.4 Bubble Point and Dew Point

At a fixed pressure:

- the **bubble-point temperature** is the temperature at which the first infinitesimal bubble of vapor appears from a liquid,
- the **dew-point temperature** is the temperature at which the first infinitesimal drop

of liquid appears from a vapor.

For a flash problem, the vapor fraction is denoted by

$$\alpha = \frac{V}{F}.$$

Then:

$$\alpha = 0 \quad \text{at the bubble point,} \quad \alpha = 1 \quad \text{at the dew point.}$$

2.5 Rachford–Rice Equation

For a multicomponent flash calculation, the standard equation is

$$\sum_i \frac{z_i(K_i - 1)}{1 + \alpha(K_i - 1)} = 0$$

where

$$K_i = \frac{y_i}{x_i}.$$

2.6 Ideal Mixture Relations for Problem B

For the ideal-mixture treatment of Problem B, the vapor phase is treated with Raoult-like equilibrium corrected by the Poynting factor:

$$K_i = \left(\frac{P_i^{sat}}{P} \right) \exp \left[\frac{V_i^L (P - P_i^{sat})}{RT} \right].$$

In the worksheet, I entered the K -value expression in CHE Calculator form as

$$K_i = (\text{Psat}(i, \text{TempBlue}) / \text{Pres}) * \text{EXP} \left(\frac{((1 / \text{Rackett}(\text{COMP ID}, \text{TempBlue})) * (\text{Pres} - \text{Psat}(\text{COMP ID}, \text{TempBlue})))}{(\text{Rg}() * (\text{TempBlue} + 273.15)))} \right) \quad (1)$$

Here, the Rackett-based term was used as an estimate of the liquid molar volume in the Poynting correction.

2.7 Why the Poynting Factor Matters

At elevated pressures, the pressure difference $(P - P_i^{sat})$ is no longer negligible, so the liquid fugacity cannot be approximated using saturation pressure alone. The exponential Poynting correction accounts for the extra work of bringing the liquid from saturation

pressure to the prevailing system pressure.

2.8 Peng–Robinson EOS Logic for Problem B

For the Peng–Robinson approach, the general calculation sequence is:

1. Specify composition, temperature, and pressure.
2. Calculate the compressibility factor Z for each phase.
3. Calculate the fugacity coefficients $\hat{\phi}_i$ of the components.
4. Construct K -values from liquid- and vapor-phase fugacity ratios.
5. Use the Rachford–Rice equation to locate bubble-point or dew-point conditions.
6. Use the converged phase compositions to compute fugacities.

2.9 Useful Numerical Strategy

This assignment is highly numerical, so I found it best to solve it in Excel by:

- creating residual equations,
- linking all temperature-dependent functions to a single working-temperature reference cell,
- using Goal Seek to drive the residual to zero by changing the guessed temperature or guessed water fraction.

A very helpful part of my worksheet setup was the use of a single cell that I called “Temp Blue.” This cell was not a separate temperature with its own physical meaning. Instead, it was a reference cell used by all equations and built-in functions that required a temperature input. Whenever a formula needed temperature, I made it point to Temp Blue rather than directly to a specific bubble-point, dew-point, or operating-temperature cell.

This made the worksheet much easier to manage. For example, the P_{sat} , K -value, EOS saturation, and other temperature-dependent expressions were all written using Temp Blue as their temperature argument. Then, depending on what I was solving, I only had to change what Temp Blue was equal to.

3. Assumptions and Given Data

3.1 General Assumptions

- Steady-state calculations.
- Thermodynamic equilibrium at phase boundaries.
- Binary interaction parameters are taken as zero in the Peng–Robinson calculations.
- In Problem A, water at saturation is in equilibrium with the natural-gas vapor phase.
- In Problem B, the ideal-mixture case retains the Poynting correction.
- In Problem B, the final state is liquid, so mixture fugacities are reported using the liquid composition.

3.2 Problem A Given Data

Table 1: Given data for Problem A

Quantity	Value
Methane	60.847 wt%
Ethane	14.256 wt%
Propane	8.363 wt%
Iso-butane	8.267 wt%
Normal-butane	8.267 wt%
Temperature	65 °F
Pressure	900 psia
EOS model	Peng–Robinson
BIPs	zero
Required output	lb/MMSCF, dew point, bubble point

3.3 Problem B Given Data

Table 2: Given data for Problem B

Quantity	Value
Component 1	Methanol
Component 2	Ethanol
Component 3	<i>n</i> -Butanol
Temperature	25 °C
Pressure	20 bar
Overall molar composition	[0.2, 0.3, 0.5]
Models required	Ideal mixture and Peng–Robinson EOS
Special note	Do not ignore Poynting factor in ideal case

4. Problem A — Natural Gas Saturation with Water

4.1 Purpose of the Problem A Calculation

The goal of Problem A was to determine the maximum amount of water that the natural gas mixture can hold at the specified operating condition. In thermodynamic language, this means finding the vapor-phase water content at which the mixture becomes saturated and is just about to form a drop of pure liquid water.

My strategy was:

1. Convert the given hydrocarbon mass fractions to dry-gas mole fractions.
2. Add water as a sixth component with an unknown small mole fraction.
3. Calculate the fugacity of water in the vapor-phase mixture using Peng–Robinson.
4. Calculate the fugacity of liquid water at the same temperature and pressure using Peng–Robinson.
5. Form a fugacity residual:

$$R_f = f_{H_2O}^V - f_{H_2O}^L$$

6. Use Goal Seek to drive the residual to zero by changing the water composition in the mixture.
7. Convert the resulting water content into lb/MMSCF of dry gas.
8. Determine the dew-point and bubble-point temperatures of the saturated stream at the same pressure.

4.2 Step A1: Convert the Given Weight Fractions to Dry-Gas Mole Fractions

A basis of 100 g of dry gas was selected. Then the mass of each component on that basis was:

Table 3: Dry-gas mass basis for Problem A

Component	Mass on 100-g basis
Methane	60.847 g
Ethane	14.256 g
Propane	8.363 g
Iso-butane	8.267 g
Normal-butane	8.267 g

The number of moles of each component was found from

$$n_i = \frac{m_i}{MW_i}.$$

Then the dry-gas mole fractions were obtained from

$$z_i^{(\text{dry})} = \frac{n_i}{\sum n_i}.$$

Using the worksheet values, the dry-gas mole fractions became approximately:

$$\begin{aligned} z_{\text{CH}_4} &= 0.799996869, \\ z_{\text{C}_2\text{H}_6} &= 0.099999760, \\ z_{\text{C}_3\text{H}_8} &= 0.040002557, \\ z_{i\text{-C}_4\text{H}_{10}} &= 0.030000407, \\ z_{n\text{-C}_4\text{H}_{10}} &= 0.030000407. \end{aligned}$$

4.3 Step A2: Enter Temperature and Pressure in Working Units

The given field conditions were:

$$T = 65^\circ\text{F}, \quad P = 900 \text{ psia}.$$

In the worksheet, these were converted using the CHE functions:

$$=\text{FtoC}(65) \quad \text{and} \quad =\text{PSIAtoBAR}(900)$$

This produced approximately:

$$T = 18.333,3^\circ\text{C}, \quad P = 62.052,6 \text{ bar}.$$

4.4 Step A3: Define the Saturated Gas Composition with Respect to Water

A water mole fraction $z_{\text{H}_2\text{O}}$ was introduced. Since the hydrocarbon mole fractions were originally defined on a dry-gas basis, the hydrocarbon fractions in the wet saturated gas had to be scaled so that the total composition still summed to one.

If $z_{\text{H}_2\text{O}}$ is the water mole fraction in the saturated gas, then the remaining hydrocarbon fraction is

$$1 - z_{\text{H}_2\text{O}}.$$

Therefore, each hydrocarbon wet-gas mole fraction was written as

$$z_i^{(\text{wet})} = (1 - z_{H_2O}) z_i^{(\text{dry})}.$$

In my spreadsheet, this is why I multiplied each dry-gas mole fraction by a factor of the form $(1 - z_{H_2O})$. That factor rescaled the dry-gas composition after water was added so that the full six-component mole-fraction set still remained normalized.

4.5 Step A4: Specify the Initial Phase Guess for the Saturation Calculation

Because the saturated natural gas at the given condition is primarily a vapor mixture, I initially set the vapor-phase composition equal to the overall composition:

$$y_i = z_i.$$

For the liquid phase, I initially set all liquid fractions equal to zero except for water, which was set equal to 1:

$$x_{H_2O} = 1, \quad x_i = 0 \text{ for hydrocarbons.}$$

This reflects the physical idea that, at the saturation limit, the incipient liquid phase is a tiny drop of *pure water*.

4.6 Step A5: Calculate the Vapor and Liquid Compressibility Factors

For the vapor phase, the mixture compressibility factor Z^V was calculated using

$$=ZComp(y_i, Temp, Pres, 1)$$

where the last input of 1 selects the vapor root. For the liquid phase, the liquid compressibility factor Z^L was calculated using

$$=ZComp(x_i, Temp, Pres, 0)$$

where the last input of 0 selects the liquid root.

4.7 Step A6: Calculate Fugacity Coefficients

After the Z -values were obtained, I calculated the component fugacity coefficients. For the vapor phase, I used

$$=PhiC(y_i, ComponentID, Temp, Pres, ZV)$$

and for the liquid phase, I used

$$= \text{PhiC}(\text{xi}, \text{ComponentID}, \text{Temp}, \text{Pres}, \text{ZL}).$$

For water, the vapor-phase fugacity was calculated as

$$f_{H_2O}^V = \hat{\phi}_{H_2O}^V P y_{H_2O}$$

and the liquid-phase fugacity was

$$f_{H_2O}^L = \hat{\phi}_{H_2O}^L P x_{H_2O}.$$

Since $x_{H_2O} = 1$, this reduces to

$$f_{H_2O}^L = \hat{\phi}_{H_2O}^L P.$$

4.8 Step A7: Fugacity Residual and Saturation Condition

The saturation criterion was enforced through the residual

$$R_f = f_{H_2O}^V - f_{H_2O}^L.$$

Then Goal Seek was used:

- **Set cell:** fugacity residual
- **To value:** 0
- **By changing cell:** water mole fraction in the wet gas

4.9 Step A8: Final Saturated Water Mole Fraction

From the worksheet, the converged water component fraction in the saturated natural gas stream was

$$z_{H_2O} = 0.0007651128$$

and the corresponding water-to-dry-gas mole ratio was

$$\frac{y_{H_2O}}{1 - y_{H_2O}} = 0.0007656986.$$

4.10 Step A9: Convert Water Content to lb/MMSCF

Using the molecular weight of water,

$$MW_{H_2O} = 18.015,$$

the moles of water per mole of dry gas were converted to water mass on a dry-gas basis:

$$\text{water mass on dry-gas basis} = \left(\frac{y_{H_2O}}{1 - y_{H_2O}} \right) MW_{H_2O}.$$

Using the worksheet value, this gave

$$0.0138 \text{ g water/gmol dry gas}$$

or

$$0.0304 \text{ lb water}$$

for the selected basis used in the sheet.

Then the dry-gas basis was converted to dry-gas volume:

$$1.0000 \text{ (chosen dry-gas basis)} \longrightarrow 0.0008 \text{ MMSCF.}$$

Finally,

$$\text{Water content in lb/MMSCF} = \frac{\text{water mass in lb}}{\text{dry-gas volume in MMSCF}}$$

which gave

$$\boxed{36.3501 \text{ lb/MMSCF}}.$$

4.11 Step A10: Bubble-Point and Dew-Point Temperatures of the Saturated Gas

After I determined the final saturated six-component gas composition, I used that composition to calculate the bubble-point and dew-point temperatures of the stream at the same pressure of 900 psia. For this part, I followed the same iterative Peng–Robinson workflow I had been using in the worksheet.

4.11.1 Bubble-Point Temperature Procedure

For the bubble-point calculation, I set

$$\alpha = 0$$

because at the bubble point the system is still essentially liquid and only the first infinitesimal bubble of vapor is about to form.

I set the liquid mole fractions equal to the overall composition:

$$x_i = z_i$$

and I set the vapor mole fractions initially to zero except for methane, which I set equal to 1 as the first incipient-vapor guess.

For the initial bubble-point temperature guess in Problem A, I used the weighted critical-temperature expression

$$T_{\text{guess,bubble}} = 0.7 \left(\sum_i z_i T_{c,i}^{(K)} \right) - 273.15$$

with all critical temperatures first expressed in Kelvin. After calculating the guess, I pasted the number itself into the bubble-temperature cell.

Then I calculated the vapor-phase compressibility factor using

$$=ZComp(y_i, TempBlue, Pres, 1)$$

and the liquid-phase compressibility factor using

$$=ZComp(x_i, TempBlue, Pres, 0).$$

Next, I calculated the fugacity coefficient of each component in each phase using

$$=PhiC(y_i, ComponentID, TempBlue, Pres, ZV)$$

for the vapor phase and

$$=PhiC(x_i, ComponentID, TempBlue, Pres, ZL)$$

for the liquid phase.

Then I formed the K-values from

$$K_i = \frac{\hat{\phi}_i^L}{\hat{\phi}_i^V}.$$

With $\alpha = 0$, I used Goal Seek to drive the Rachford–Rice residual to zero by changing the bubble-point temperature cell.

That first pass only brought the solution close to the true bubble point because the original vapor-phase guess assumed that only methane was vaporizing. After that first pass, I updated the vapor composition using

$$y_i = K_i x_i$$

and pasted those new vapor fractions back into the worksheet as numbers. Then I Goal Sought the Rachford–Rice equation again. To finish the calculation cleanly, I used the

built-in CHE Calculator convergence routine.

Using that procedure, the final bubble-point temperature was

$$T_{\text{bubble}} = -6.23561012^{\circ}\text{F}.$$

4.11.2 Dew-Point Temperature Procedure

For the dew-point calculation, I set

$$\alpha = 1$$

because at the dew point the system is still essentially all vapor and only the first infinitesimal drop of liquid is about to form.

I set the vapor mole fractions equal to the overall composition:

$$y_i = z_i$$

and I set the liquid mole fractions initially to zero except for water, which I set equal to 1 as the first incipient-liquid guess.

For the initial dew-point temperature guess in Problem A, I used

$$T_{\text{guess,dew}} = 1.1 \left(\sum_i z_i T_{c,i}^{(\text{K})} \right) - 273.15.$$

Then I calculated the vapor-phase compressibility factor from

$$= \text{ZComp}(y_i, \text{TempBlue}, \text{Pres}, 1)$$

and the liquid-phase compressibility factor from

$$= \text{ZComp}(x_i, \text{TempBlue}, \text{Pres}, 0).$$

After that, I calculated the fugacity coefficient of each component in the vapor phase using

$$= \text{PhiC}(y_i, \text{ComponentID}, \text{TempBlue}, \text{Pres}, \text{ZV})$$

and in the liquid phase using

$$= \text{PhiC}(x_i, \text{ComponentID}, \text{TempBlue}, \text{Pres}, \text{ZL}).$$

Then I formed the K-values from

$$K_i = \frac{\hat{\phi}_i^L}{\hat{\phi}_i^V}.$$

With $\alpha = 1$, I used Goal Seek to drive the Rachford–Rice residual to zero by changing the dew-point temperature cell.

After that first pass, I updated the liquid composition from

$$x_i = \frac{y_i}{K_i}$$

and pasted those new liquid fractions back into the worksheet as numbers. Then I Goal Sought the Rachford–Rice equation again. To finish the calculation, I used the built-in CHE Calculator convergence routine.

Using that procedure, the final dew-point temperature was

$$T_{\text{dew}} = 65.00111298^\circ\text{F}.$$

4.12 Reference Check Using Saturation Pressures

As an additional check, I also calculated the saturation pressure of each component at the given temperature using

$$=\text{EOSPsat}(\text{ComponentID}, \text{TempBlue}).$$

The bubble-point pressure was calculated from

$$P_{\text{bubble}} = \sum_i z_i P_i^{\text{sat}}$$

which in Excel was entered as

$$=\text{SUMPRODUCT}(z_i, C7:H7).$$

Using the saturated-stream composition and the saturation pressures evaluated at the working temperature guess, this gave

$$P_{\text{bubble}} = 243.7878.$$

The dew-point pressure was calculated from

$$P_{\text{dew}} = \frac{1}{\sum_i \frac{z_i}{P_i^{\text{sat}}}}$$

which in Excel was entered as

$$=1/\text{SUMPRODUCT}(z_i, 1/C7:H7).$$

Using the same composition and saturation-pressure values, this gave

$$P_{\text{dew}} = 12.5801.$$

4.12.1 Interpretation of the Final Bubble and Dew Temperatures

The final values obtained were

$$T_{\text{bubble}} = -6.23561012^\circ\text{F}$$

and

$$T_{\text{dew}} = 65.00111298^\circ\text{F}.$$

These results are physically consistent. The actual stream temperature is

$$65^\circ\text{F},$$

which is essentially equal to the dew-point temperature. That confirms that the gas is saturated with water at the specified condition. The bubble-point temperature lies far below the operating temperature, which is also reasonable because the stream at these conditions is predominantly vapor rather than liquid.

4.13 Interpretation of the Problem A Results

The final water content of 36.3501 lb/MMSCF means that at 900 psia and 65°F, the gas can hold approximately 36.35 pounds of water for every million standard cubic feet of dry gas before free liquid water begins to form.

The dew-point result, $T_{\text{dew}} \approx 65^\circ\text{F}$, is especially important because it tells me that the stream is right at the onset of water condensation at the specified condition.

4.14 Problem A Final Answers

Final Result Box

Maximum water content = 36.3501 lb/MMSCF

$$T_{\text{bubble}} = -6.2356\text{ }^{\circ}\text{F}$$

$$T_{\text{dew}} = 65.0011\text{ }^{\circ}\text{F}$$

4.15 Excel Screenshots for Problem A

The screenshots below document the actual spreadsheet setup used for the natural-gas saturation calculation.

VLE Calculations		Peng-Robinson						User Calculations				
Fluid Conditions		Mixture	CH4	C2H6	C3H8	C4H10	C4H10	H2O				
Molar Composition		1.0000	0.799385	0.099923	0.039972	0.029977	0.029977	0.000765				
Temperature, C		18.3333										
Pressure, bar		62.0526										
		18.3333										
Saturated Vapor Pressure, bar		0.2401	0.1093	0.0540	0.0091	2.9047	1.9975	0.0175				
Bubble Point Pressure, bar		0.0734										
Dew Point Pressure, bar												
Saturated Vapor Temperature, C												
Bubble Point Temperature, C		-21.2420										
Dew Point Temperature, C		18.3340										
K-Values			26624.5947	38928.6020	573287.4101	28196871.78	9693827.69	0.0008				
New K-Values												
Vapor to Feed Molar Ratio ($\alpha=V/F$)												
The Rachford-Rice Equation												
Vapor Phase		Mixture										
Molar Composition		1.0000	0.799385	0.099923	0.039972	0.029977	0.029977	0.000765				
New Molar Composition		0.9990	0.798724	0.099811	0.039910	0.029917	0.029918	0.000765				
Compressibility Factor		0.7492										
Fugacity Coefficient			0.8873	0.5867	0.4166	0.3174	0.2969	0.3886				
Saturated Fugacity Coefficient												
Molar Enthalpy, kJ/kmol												
Molar Entropy, kJ/(kmol K)												
Molar Density, kmol/m ³												
Molar Heat Capacity, kJ/(kmol K)												
Liquid Phase		Mixture										
Molar Composition		1.0000	0.000030	0.000003	0.000000	0.000000	0.000000	0.999967				
New Molar Composition		1.0000	0.000030	0.000003	0.000000	0.000000	0.000000	1.000000				
Compressibility Factor		0.0541										
Fugacity Coefficient			23624.4788	22840.3842	238818.1258	8948900.4783	2878109.5311	0.0003				
Liquid Activity Coefficient												
Molar Enthalpy, kJ/kmol												
Molar Entropy, kJ/(kmol K)												
Molar Density, kmol/m ³												
Molar Heat Capacity, kJ/(kmol K)												
Molar Heat of Vaporization, kJ/kmol												

Figure 1: Peng–Robinson VLE worksheet used for the saturated natural-gas stream.

User Data and Calculations: ON						
	SUMS					
Weight Fractions	100.0000	60.847000000	14.256000000	8.363000000	8.267000000	8.267000000
Moles with Basis of 100 gmols	4.7409	3.792744499	0.474093781	0.18965009	0.142230404	0.142230404
Mole Fractions	1.0000	0.799996869	0.099999760	0.040002557	0.030000407	0.030000407
Fugacity of Water (V)		0.0184				
Fugacity of Water (L)		0.0184				
Fugacity of Water Residual		0.0000				
Water Component Fraction (Z_H2O)		0.0007651128				
Water-to-Dry-Gas Mole Ratio [$y_{H2O}/(1-y_{H2O})$]		0.0007656986				
Water Molecular Weight		18.0150				
Water Mass on Dry-Gas Basis		0.0138				
Water Mass in lb		0.0304				
Chosen Dry-Gas Basis		1.0000				
Dry-Gas Volume in MMSCF		0.0008				
Water Content in lb/MMSCF		36.3501				

Figure 2: User-calculation section for Problem A showing the conversion from dry-gas weight fractions to mole fractions and the final lb/MMSCF value.

5. Problem B — Fugacity Calculations for Methanol / Ethanol / Normal Butanol

5.1 Purpose of the Problem B Calculation

Problem B required two parallel analyses of the same mixture:

1. an **ideal-mixture** treatment with the Poynting factor included,
2. a **Peng–Robinson** EOS treatment.

For each model, the first step was to calculate the bubble-point and dew-point temperatures to verify whether the mixture is liquid, vapor, or two-phase at the actual operating condition of 25 °C and 20 bar. After the phase region was identified, the next task was to calculate the fugacities of the pure components and the component fugacities in the mixture.

5.2 Overall Composition

The overall molar composition was

$$z_1 = 0.2, \quad z_2 = 0.3, \quad z_3 = 0.5$$

for methanol, ethanol, and normal butanol, respectively.

5.3 Step B1: Create a Single Working Temperature Cell

To make Goal Seek easier, I used a single common temperature cell for all temperature-dependent property calculations. In my worksheet, I called this the “Temp Blue” cell. This setup was useful because all P_{sat} , K -value, and EOS-based temperature-dependent functions referenced the same cell.

5.4 Ideal-Mixture Model

5.4.1 Step B2: Calculate Saturation Temperatures for the Initial Guess

For the ideal-mixture part, I used the saturated temperatures at the system pressure as the main guide for the initial bubble-point and dew-point guesses. In the worksheet, these were calculated from the corresponding saturation-temperature function at the specified pressure. These values gave me a reasonable temperature range before I started the Goal Seek procedure.

5.4.2 Step B3: Include the Poynting Factor in the K-Values

The ideal-mixture K-values were not taken simply as P_i^{sat}/P . Instead, the Poynting correction was included:

$$K_i = \left(\frac{P_i^{sat}}{P} \right) \exp \left[\frac{V_i^L (P - P_i^{sat})}{RT} \right].$$

In the Excel setup this was implemented as

$$=(D7/Pres)*EXP(((1/Rackett(D2,TempBlue))*(Pres-D7))/(Rg()*(TempBlue+273.15)))$$

5.4.3 Step B4: Bubble-Point Temperature for the Ideal Case

For the bubble-point temperature, I set

$$\alpha = 0$$

and Goal Sought the bubble-point form of the Rachford–Rice equation to zero by changing the bubble-temperature cell.

The ideal-mixture bubble-point temperature obtained was

$$T_{\text{bubble,ideal}} = 210.9351785^\circ\text{C}.$$

5.4.4 Step B5: Dew-Point Temperature for the Ideal Case

For the dew-point temperature, I set

$$\alpha = 1$$

and Goal Sought the dew-point form of the Rachford–Rice equation to zero by changing the dew-temperature cell.

The ideal-mixture dew-point temperature obtained was

$$T_{\text{dew,ideal}} = 231.4970851^\circ\text{C}.$$

5.4.5 Step B6: Verify the Phase Region for the Ideal Case

Since

$$25^\circ\text{C} < 210.9352^\circ\text{C} < 231.4971^\circ\text{C},$$

the mixture lies well below the bubble-point temperature at the prevailing pressure. Therefore, the mixture is in the liquid phase, so for the ideal-mixture fugacity calculation I

used

$$x_i = z_i.$$

5.4.6 Step B7: Pure-Component Fugacities for the Ideal Case

For the ideal model, the pure-component liquid fugacity values obtained in the worksheet were

$$\begin{aligned}f_1^{\text{pure, ideal}} &= 0.1587 \text{ bar} && \text{(methanol)} \\f_2^{\text{pure, ideal}} &= 0.0729 \text{ bar} && \text{(ethanol)} \\f_3^{\text{pure, ideal}} &= 0.0101 \text{ bar} && \text{(normal butanol)}.\end{aligned}$$

5.4.7 Step B8: Component Fugacities in the Mixture for the Ideal Case

The component fugacities inside the ideal liquid mixture were

$$\begin{aligned}f_1^{\text{mix, ideal}} &= 0.0317 \text{ bar} \\f_2^{\text{mix, ideal}} &= 0.0219 \text{ bar} \\f_3^{\text{mix, ideal}} &= 0.0050 \text{ bar}.\end{aligned}$$

5.5 Peng–Robinson EOS Model

5.5.1 Step B9: EOS Saturation Functions for Guidance

In addition to the main VLE calculation, I also used EOS-based saturated properties as reference checks. For each component, I calculated the EOS-based saturation pressure using

$$= \text{EOSPsat}(\text{ComponentID}, \text{TempBlue})$$

and the EOS-based saturation temperature at the system pressure using

$$= \text{EOSTsat}(\text{ComponentID}, \text{Pres}).$$

For the Peng–Robinson case, I used the EOS saturation temperatures together with the weighted critical-temperature guess formulas to choose initial values for the bubble-point and dew-point temperatures.

5.5.2 Step B10: Bubble-Point Temperature for the Peng–Robinson Case

For the Peng–Robinson bubble-point calculation, I followed the same general logic as in Problem A.

At the bubble point, the system is still essentially all liquid and only the first infinitesimal

bubble of vapor is about to form, so I set

$$\alpha = 0.$$

I set the liquid fractions equal to the overall composition,

$$x_i = z_i,$$

and I set the vapor fractions initially to zero except for the lightest component, methanol, which I set equal to 1 as the first incipient-vapor guess.

For the initial bubble-point temperature guess in the Peng–Robinson case, I used

$$T_{\text{guess,bubble}} = 0.7 \left(\sum_i z_i T_{c,i}^{(K)} \right) - 273.15.$$

Then I calculated the vapor-phase compressibility factor using

$$=ZComp(yi, TempBlue, Pres, 1)$$

and the liquid-phase compressibility factor using

$$=ZComp(xi, TempBlue, Pres, 0).$$

Once those Z-values were available, I calculated the fugacity coefficient of each component in each phase. For the vapor phase, I used

$$=PhiC(yi, ComponentID, TempBlue, Pres, ZV)$$

and for the liquid phase, I used

$$=PhiC(xi, ComponentID, TempBlue, Pres, ZL).$$

Then I formed the K-values from

$$K_i = \frac{\hat{\phi}_i^L}{\hat{\phi}_i^V}.$$

With $\alpha = 0$, I used Goal Seek to drive the Rachford–Rice residual to zero by changing the bubble-point temperature cell.

That first pass only brought the calculation close to the true bubble point because the original vapor guess assumed that only methanol was present in the incipient vapor phase.

After that first solve, I updated the vapor composition using

$$y_i = K_i x_i$$

and pasted those updated vapor fractions back into the worksheet as numbers. Then I repeated the Goal Seek step. To finish the calculation, I used the built-in CHE Calculator convergence routine.

Using that procedure, the final Peng–Robinson bubble-point temperature was

$$T_{\text{bubble,PR}} = 198.95278^\circ\text{C}.$$

As an additional pressure check, the bubble-point pressure at the actual temperature of 25 °C was

$$P_{\text{bubble,PR}} = 0.0632 \text{ bar}.$$

5.5.3 Step B11: Dew-Point Temperature for the Peng–Robinson Case

For the Peng–Robinson dew-point calculation, I used the same procedure from the vapor side.

At the dew point, the system is still essentially all vapor and only the first infinitesimal drop of liquid is about to form, so I set

$$\alpha = 1.$$

I set the vapor fractions equal to the overall composition,

$$y_i = z_i,$$

and I set the liquid fractions initially to zero except for the heaviest component, *n*-butanol, which I set equal to 1 as the first incipient-liquid guess.

For the initial dew-point temperature guess in the Peng–Robinson case, I used

$$T_{\text{guess,dew}} = 1.1 \left(\sum_i z_i T_{c,i}^{(\text{K})} \right) - 273.15.$$

Then I calculated the vapor-phase compressibility factor from

$$=ZComp(y_i, TempBlue, Pres, 1)$$

and the liquid-phase compressibility factor from

$$=ZComp(xi, TempBlue, Pres, 0).$$

After that, I calculated the fugacity coefficient of each component in the vapor phase using

$$=PhiC(yi, ComponentID, TempBlue, Pres, ZV)$$

and in the liquid phase using

$$=PhiC(xi, ComponentID, TempBlue, Pres, ZL).$$

Then I formed the K-values from

$$K_i = \frac{\hat{\phi}_i^L}{\hat{\phi}_i^V}.$$

With $\alpha = 1$, I used Goal Seek to drive the Rachford–Rice residual to zero by changing the dew-point temperature cell.

After that first pass, I updated the liquid composition from

$$x_i = \frac{y_i}{K_i}$$

and pasted those updated liquid fractions back into the worksheet as numbers. Then I repeated the Goal Seek step. To finish the calculation, I used the built-in CHE Calculator convergence routine.

Using that procedure, the final Peng–Robinson dew-point temperature was

$$T_{\text{dew,PR}} = 214.330973^\circ\text{C}.$$

As an additional pressure check, the dew-point pressure at the actual temperature of 25°C was

$$P_{\text{dew,PR}} = 0.0249 \text{ bar}.$$

5.5.4 Step B12: Verify the Phase Region for the Peng–Robinson Case

To determine the phase region, I compared the actual system temperature to the Peng–Robinson bubble-point and dew-point temperatures:

$$25^\circ\text{C} < 198.9528^\circ\text{C} < 214.3310^\circ\text{C}.$$

This shows that the actual temperature is far below the bubble-point temperature. Therefore,

the mixture is in the liquid phase under the Peng–Robinson model.

This conclusion is also consistent with the pressure checks at the actual temperature:

$$P_{\text{bubble,PR}} = 0.0632 \text{ bar} \quad \text{and} \quad P_{\text{dew,PR}} = 0.0249 \text{ bar}.$$

Since the actual system pressure is

$$P = 20 \text{ bar},$$

which is much greater than both of these pressures, the mixture is confirmed to be liquid at 25 °C under the Peng–Robinson model. So in the final Peng–Robinson fugacity calculation, I used

$$x_i = z_i.$$

5.5.5 Step B13: Why the Liquid-Phase Fugacity Formula Looks Like $P \times \phi$

Once the liquid composition was known, the component fugacity in the mixture was obtained by multiplying the fugacity coefficient by the pressure and by the component mole fraction:

$$f_i^{\text{mix}} = \hat{\phi}_i x_i P.$$

If the spreadsheet contained something like

$$=\text{Pres}*\text{Phi},$$

that by itself represented the fugacity of the pure liquid phase at composition 1 for that component. Once it was multiplied by the component mole fraction,

$$=\text{component mole fraction}*\text{Pres}*\text{Phi},$$

the result became the fugacity of that component inside the mixture.

5.5.6 Step B14: Pure-Component Fugacities for the Peng–Robinson Case

To calculate the pure-component fugacity of each component with the Peng–Robinson model, I temporarily changed the liquid composition so that the component I was interested in had liquid mole fraction equal to 1, while the other liquid mole fractions were set to 0. In other words, I treated that component as the pure liquid phase at the specified temperature and pressure, recalculated the liquid-phase fugacity coefficient, and then calculated the pure fugacity from

$$f_i^{\text{pure}} = \hat{\phi}_i P.$$

Using that procedure, the pure-component fugacities obtained from the Peng–Robinson

worksheet were

$$\begin{aligned} f_1^{\text{pure, PR}} &= 0.1648 \text{ bar} && (\text{methanol}) \\ f_2^{\text{pure, PR}} &= 0.0847 \text{ bar} && (\text{ethanol}) \\ f_3^{\text{pure, PR}} &= 0.0153 \text{ bar} && (\text{normal butanol}). \end{aligned}$$

5.5.7 Step B15: Component Fugacities in the Mixture for the Peng–Robinson Case

The Peng–Robinson component fugacities in the liquid mixture were

$$\begin{aligned} f_1^{\text{mix, PR}} &= 0.0393 \text{ bar} \\ f_2^{\text{mix, PR}} &= 0.0266 \text{ bar} \\ f_3^{\text{mix, PR}} &= 0.0080 \text{ bar}. \end{aligned}$$

5.6 Step B16: Final Phase-Verification Summary

Both models gave phase-boundary temperatures far above the actual temperature:

$$T_{\text{actual}} = 25^\circ\text{C}.$$

Therefore, the mixture is liquid under both the ideal-mixture model and the Peng–Robinson model. That is why the final fugacity calculations were carried out using the liquid composition.

5.7 Problem B Final Results Table

Table 4: Problem B fugacity results

Component	$f^{\text{pure, ideal}}$ (bar)	$f^{\text{mix, ideal}}$ (bar)	$f^{\text{pure, PR}}$ (bar)	$f^{\text{mix, PR}}$ (bar)
Methanol	0.1587	0.0317	0.1648	0.0393
Ethanol	0.0729	0.0219	0.0847	0.0266
Normal butanol	0.0101	0.0050	0.0153	0.0080

Table 5: Problem B phase-boundary temperatures

Model	Bubble-point temperature	Dew-point temperature
Ideal mixture	210.9352°C	231.4971°C
Peng–Robinson	198.9528°C	214.3310°C

5.8 Problem B Final Answers

Final Result Box

$$T_{\text{bubble, ideal}} = 210.9352^{\circ}\text{C}$$

$$T_{\text{dew, ideal}} = 231.4971^{\circ}\text{C}$$

$$T_{\text{bubble, PR}} = 198.9528^{\circ}\text{C}$$

$$T_{\text{dew, PR}} = 214.3310^{\circ}\text{C}$$

$$f_1^{\text{pure, ideal}} = 0.1587 \text{ bar}$$

$$f_1^{\text{mix, ideal}} = 0.0317 \text{ bar}$$

$$f_2^{\text{pure, ideal}} = 0.0729 \text{ bar}$$

$$f_2^{\text{mix, ideal}} = 0.0219 \text{ bar}$$

$$f_3^{\text{pure, ideal}} = 0.0101 \text{ bar}$$

$$f_3^{\text{mix, ideal}} = 0.0050 \text{ bar}$$

$$f_1^{\text{pure, PR}} = 0.1648 \text{ bar}$$

$$f_1^{\text{mix, PR}} = 0.0393 \text{ bar}$$

$$f_2^{\text{pure, PR}} = 0.0847 \text{ bar}$$

$$f_2^{\text{mix, PR}} = 0.0266 \text{ bar}$$

$$f_3^{\text{pure, PR}} = 0.0153 \text{ bar}$$

$$f_3^{\text{mix, PR}} = 0.0080 \text{ bar}$$

The mixture is liquid at 25°C under both models

5.9 Excel Screenshots for Problem B

The first worksheet below shows the ideal-mixture model with the Poynting factor retained. The second worksheet shows the Peng–Robinson EOS results. Together, they document the bubble-point and dew-point calculations and the final pure-component and mixture fugacity values.

	A	B	C	D	E	M	N	O	P	Q
1	VLE Calculations		Ideal Gas			User Calculations				
2	Fluid Conditions		Mixture	CH4O	C2H6O	C4H10O				
3	Molar Composition		1.0000	0.200000	0.300000	0.500000				
4	Temperature, C		25.0000							
5	Pressure, bar		20.0000							
6			25.0000							
7	Saturated Vapor Pressure, bar			0.1587	0.0729	0.0100				
8	Bubble Point Pressure, bar		0.0586							
9	Dew Point Pressure, bar		0.0181							
10										
11	Saturated Vapor Temperature, C			173.9812	194.7567	260.3553				
12	Bubble Point Temperature, C		210.9352							
13	Dew Point Temperature, C		231.4971							
14										
15	K-Values			0.0079	0.0036	0.0005				
16	New K-Values									
17										
18	Vapor to Feed Molar Ratio ($\alpha=V/F$)									
19	The Rachford-Rice Equation									
20	Vapor Phase	Mixture								
21	Molar Composition									
22	New Molar Composition									
23	Compressibility Factor									
24	Fugacity Coefficient									
25	Saturated Fugacity Coefficient									
26										
27	Molar Enthalpy, kJ/kmol									
28	Molar Entropy, kJ/(kmol K)									
29	Molar Density, kmol/m ³									
30	Molar Heat Capacity, kJ/(kmol K)									
31	Liquid Phase	Mixture								
32	Molar Composition		1.0000	0.200000	0.300000	0.500000				
33	New Molar Composition									
34	Compressibility Factor									
35	Fugacity Coefficient			0.1587	0.0729	0.0101				
36	Liquid Activity Coefficient									
37										
38	Molar Enthalpy, kJ/kmol									
39	Molar Entropy, kJ/(kmol K)									
40	Molar Density, kmol/m ³									
41	Molar Heat Capacity, kJ/(kmol K)									
42	Molar Heat of Vaporization, kJ/kmol									

Figure 3: Ideal-mixture worksheet for Problem B, including saturated vapor pressures, K-values with the Poynting factor, dew-point and bubble-point temperatures, and ideal fugacity results.

	A	B	C	D	E	M	N	O	P	Q
1	VLE Calculations		Peng-Robinson			User Calculations				
2	Fluid Conditions	Mixture	CH4O	C2H6O	C4H10O					
3	Molar Composition	1.0000	0.200000	0.300000	0.500000					
4	Temperature, C	25.0000								
5	Pressure, bar	20.0000								
6		25.0000								
7	Saturated Vapor Pressure, bar		0.1592	0.0809	0.0142					
8	Bubble Point Pressure, bar	0.0632								
9	Dew Point Pressure, bar	0.0249								
10										
11	Saturated Vapor Temperature, C		164.5350	180.9654	240.6799					
12	Bubble Point Temperature, C	198.9528								
13	Dew Point Temperature, C	214.3310								
14										
15	K-Values		1.0788	1.0370	0.9164					
16	New K-Values									
17										
18	Vapor to Feed Molar Ratio [$\alpha=V/F$]									
19	The Rachford-Rice Equation									
20	Vapor Phase	Mixture								
21	Molar Composition	1.0000	0.332695	0.389450	0.277855					
22	New Molar Composition	0.9851	0.215766	0.311100	0.458224					
23	Compressibility Factor	0.0528								
24	Fugacity Coefficient		0.0091	0.0043	0.0009					
25	Saturated Fugacity Coefficient									
26										
27	Molar Enthalpy, kJ/kmol									
28	Molar Entropy, kJ/(kmol K)									
29	Molar Density, kmol/m ³									
30	Molar Heat Capacity, kJ/(kmol K)									
31	Liquid Phase	Mixture								
32	Molar Composition	1.0000	0.200000	0.300000	0.500000					
33	New Molar Composition	0.9871	0.308384	0.375555	0.303187					
34	Compressibility Factor	0.0600								
35	Fugacity Coefficient		0.0098	0.0044	0.0008					
36	Liquid Activity Coefficient									
37		PHI*PRES	0.1967	0.0886	0.0160					
38	Molar Enthalpy, kJ/kmol									
39	Molar Entropy, kJ/(kmol K)	PHI*PRES*x _i	0.0393	0.0266	0.0080					
40	Molar Density, kmol/m ³									
41	Molar Heat Capacity, kJ/(kmol K)									
42	Molar Heat of Vaporization, kJ/kmol									

Figure 4: Peng–Robinson worksheet for Problem B, showing dew-point and bubble-point temperatures, liquid-phase composition handling, fugacity coefficients, and pure/mixture fugacity values.

5.10 Excel Fugacity Table Screenshot for Problem B

	Ideal Solution		Non-Ideal Solution	
	Pure Component	Component in Mixture	Pure Component	Component in Mixture
Methanol	0.158731469	0.031746294	0.164750277	0.039342472
Ethanol	0.072911438	0.021873431	0.084654502	0.026578789
N-Butanol	0.010050142	0.005025071	0.015277538	0.007998906

Figure 5: Focused Excel screenshot showing the fugacity table used for Problem B. This image highlights the worksheet section where the pure-component fugacities and the component fugacities in the mixture were recorded.

6. Discussion

6.1 Discussion of Problem A

Problem A is a very nice engineering example because the final number is reported in an industry unit, lb/MMSCF, but the underlying thermodynamic condition is a strict phase-equilibrium requirement. The final water content of 36.3501 lb/MMSCF was found by equating the vapor-phase water fugacity to the liquid-water fugacity. That is the correct real-fluid statement of the condition that the gas is saturated with water.

The dew-point result of approximately 65.0011°F is especially revealing, since it matches the actual stream temperature almost exactly. That confirms the physical consistency of the calculation: the gas was solved precisely at the point where it becomes saturated. The bubble-point temperature of -6.2356°F is much lower, which is also consistent with the fact that the stream is overwhelmingly vapor rather than liquid at the prevailing conditions.

6.2 Discussion of Problem B

Problem B shows the difference between ideal-mixture and real-mixture treatments. Both models predict very high bubble-point and dew-point temperatures compared with the operating temperature of 25°C, so both models agree that the mixture is in the liquid phase. However, the actual fugacity values differ between the ideal and PR treatments, which reflects the non-ideal behavior captured by the equation of state.

The Peng–Robinson model predicts slightly different pure and mixture fugacities for all three alcohols. That is expected because the EOS accounts for real-fluid effects more directly than the ideal-mixture model. Even so, both models preserve the same physical phase conclusion.

6.3 Why the Excel Structure Was Helpful

This assignment involved several linked calculations, including composition conversion, phase-composition updates, fugacity-coefficient calculations, and residual equations for saturation, bubble-point, and dew-point conditions. Excel was especially useful because each step could be built clearly and checked directly, and Goal Seek could then be used to drive the governing equations to zero.

7. Conclusion

This report presented a full thermodynamic solution strategy for CHE 301 Assignment #06. In Problem A, the natural gas composition was first converted from weight fractions to mole fractions and then combined with water as an additional component. Using the Peng–Robinson EOS, the maximum water content was determined by solving the condition

$$f_{H_2O}^V = f_{H_2O}^L.$$

The final result was 36.3501 lb/MMSCF, with a bubble-point temperature of -6.2356°F and a dew-point temperature of 65.0011°F .

In Problem B, the methanol/ethanol/*n*-butanol mixture was analyzed first as an ideal liquid mixture with the Poynting correction and then as a real mixture using the Peng–Robinson EOS. In both cases, the bubble-point and dew-point temperatures were far above the operating temperature of 25°C , confirming that the mixture is liquid at the specified condition. The fugacities of the pure components and of the components in the mixture were then calculated and summarized in tables.

Overall, the results are consistent with the governing thermodynamic theory and with the numerical procedures used in CHE Calculator and Excel.

References

1. Caracotsios, M. *An Introduction to Chemical Engineering Thermodynamics: Theory and Applications*, First Edition.
 - Chapter 4: Equations of State for Pure Fluids
 - Chapter 5: Thermal Properties of Homogeneous Mixtures
 - Chapter 6: Equations of State for Non-Ideal Gas and Liquid Mixtures
2. CHE 301 Assignment #06 handout.
3. Class notes and CHE Calculator worksheet functions used in the Excel implementation.

Statement on Use of External Tools

External tools were used for organizing equations, improving formatting, and helping structure the written report. The thermodynamic setup, Excel workflow, interpretation of the CHE Calculator output, and the final review of the calculations were completed by me.

8. Useful Equations Summary

Problem A

$$z_i^{(\text{wet})} = (1 - z_{\text{H}_2\text{O}}) z_i^{(\text{dry})}$$

$$f_{\text{H}_2\text{O}}^V = \hat{\phi}_{\text{H}_2\text{O}}^V P y_{\text{H}_2\text{O}}$$

$$f_{\text{H}_2\text{O}}^L = \hat{\phi}_{\text{H}_2\text{O}}^L P x_{\text{H}_2\text{O}}$$

$$f_{\text{H}_2\text{O}}^V = f_{\text{H}_2\text{O}}^L$$

$$\frac{y_{\text{H}_2\text{O}}}{1 - y_{\text{H}_2\text{O}}} \quad (\text{water-to-dry-gas mole ratio})$$

Problem B

$$K_i = \frac{y_i}{x_i}$$

$$\sum_i \frac{z_i(K_i - 1)}{1 + \alpha(K_i - 1)} = 0$$

$$\alpha = 0 \quad (\text{bubble point}) \quad \alpha = 1 \quad (\text{dew point})$$

$$f_i^{\text{mix}} = \hat{\phi}_i x_i P$$

$$f_i^{\text{pure}} = \hat{\phi}_i P$$

A. Excel Checklist for Problem A

Problem A Excel Checklist — Part 1 of 3

1. Enter the dry-gas weight fractions for methane, ethane, propane, iso-butane, and normal-butane.
2. Choose a dry-gas basis, such as 100 g, so the mass-to-mole conversion is easy to organize.
3. Convert each hydrocarbon from mass to moles using

$$n_i = \frac{m_i}{MW_i},$$

then calculate the dry-gas mole fractions from

$$z_i^{(\text{dry})} = \frac{n_i}{\sum n_i}.$$

4. Convert the given temperature and pressure into worksheet units using

$$=FtoC(65) \quad \text{and} \quad =PSIAtoBAR(900).$$

5. Add water as a sixth component with unknown mole fraction z_{H_2O} .
6. Define the wet-gas composition from

$$z_i^{(\text{wet})} = (1 - z_{H_2O}) z_i^{(\text{dry})}.$$

7. Set the vapor-phase composition equal to the wet-gas composition.

Problem A Excel Checklist — Part 2 of 3

8. Set the liquid-phase guess to pure water:

$$x_{H_2O} = 1, \quad x_i = 0 \text{ for hydrocarbons.}$$

9. Calculate the vapor and liquid compressibility factors using

$$=ZComp(yi, Temp, Pres, 1) \quad \text{and} \quad =ZComp(xi, Temp, Pres, 0).$$

10. Calculate the vapor- and liquid-phase fugacity coefficients using

$$=PhiC(yi, ComponentID, Temp, Pres, ZV)$$

$$=PhiC(xi, ComponentID, Temp, Pres, ZL)$$

11. Calculate the water fugacities:

$$f_{H_2O}^V = \hat{\phi}_{H_2O}^V P y_{H_2O}, \quad f_{H_2O}^L = \hat{\phi}_{H_2O}^L P x_{H_2O}.$$

12. Form the fugacity residual

$$R_f = f_{H_2O}^V - f_{H_2O}^L$$

and use Goal Seek to make it zero by changing z_{H_2O} .

13. After convergence, calculate the water-to-dry-gas mole ratio

$$\frac{y_{H_2O}}{1 - y_{H_2O}}$$

and convert the final result to lb/MMSCF.

14. For the bubble-point calculation, set

$$\alpha = 0, \quad x_i = z_i,$$

and use methane as the first vapor guess.

Problem A Excel Checklist — Part 3 of 3

15. Use the initial bubble-point temperature guess

$$T_{\text{guess,bubble}} = 0.7 \left(\sum_i z_i T_{c,i}^{(K)} \right) - 273.15.$$

16. Calculate Z^V , Z^L , fugacity coefficients, and K-values:

$$K_i = \frac{\hat{\phi}_i^L}{\hat{\phi}_i^V}.$$

17. Goal Seek the Rachford–Rice residual to zero by changing the bubble-point temperature. Then update the vapor composition from

$$y_i = K_i x_i$$

and repeat, or use the built-in CHE convergence routine.

18. For the dew-point calculation, set

$$\alpha = 1, \quad y_i = z_i,$$

and use water as the first liquid guess. Use the initial dew-point temperature guess

$$T_{\text{guess,dew}} = 1.1 \left(\sum_i z_i T_{c,i}^{(K)} \right) - 273.15.$$

19. Recalculate Z^V , Z^L , fugacity coefficients, and K-values. Goal Seek the Rachford–Rice residual to zero by changing the dew-point temperature, then update the liquid composition from

$$x_i = \frac{y_i}{K_i}$$

and repeat, or use the built-in CHE convergence routine. As a final check, calculate component saturation pressures with

$$=EOSPsat(\text{Component ID}, \text{Temp})$$

and use them to estimate

$$P_{\text{bubble}} = \sum_i z_i P_i^{\text{sat}}, \quad P_{\text{dew}} = \frac{1}{\sum_i z_i / P_i^{\text{sat}}}.$$

B. Excel Checklist for Problem B

Problem B Excel Checklist — Part 1 of 4

1. Enter the system conditions and overall composition:

$$T = 25^{\circ}\text{C}, \quad P = 20 \text{ bar}, \quad \mathbf{z} = [0.2, 0.3, 0.5].$$

2. Create one common working temperature cell, called “Temp Blue,” and make all temperature-dependent formulas reference that cell.
3. For the ideal case, use saturated temperatures at the system pressure as the main guide for the initial bubble-point and dew-point guesses.
4. Calculate the pure-component saturation pressures using

$$=\text{Psat}(\text{ComponentID}, \text{Temp}).$$

5. Calculate the ideal K-values using the Poynting-corrected expression.
6. For the ideal bubble-point calculation, set

$$\alpha = 0, \quad x_i = z_i,$$

then Goal Seek the Rachford–Rice residual to zero by changing the bubble-temperature cell.

Problem B Excel Checklist — Part 2 of 4

7. For the ideal dew-point calculation, set

$$\alpha = 1, \quad y_i = z_i,$$

then Goal Seek the dew-point form of the Rachford–Rice equation to zero by changing the dew-temperature cell.

8. Compare the actual temperature with the ideal bubble-point and dew-point temperatures to identify the phase region.
9. Since the actual temperature is far below the phase-boundary temperatures, use

$$x_i = z_i$$

for the final ideal-mixture fugacity calculation.

10. For the Peng–Robinson case, calculate EOS-based saturation pressures and temperatures using

$$=EOSP_{\text{sat}}(\text{ComponentID}, \text{Temp}) \quad \text{and} \quad =EOST_{\text{sat}}(\text{ComponentID}, \text{Pres}).$$

11. Use the initial Peng–Robinson temperature guesses

$$T_{\text{guess, bubble}} = 0.7 \left(\sum_i z_i T_{c,i}^{(\text{K})} \right) - 273.15$$

and

$$T_{\text{guess, dew}} = 1.1 \left(\sum_i z_i T_{c,i}^{(\text{K})} \right) - 273.15.$$

12. For the PR bubble-point calculation, set

$$\alpha = 0, \quad x_i = z_i,$$

and use the lightest component as the first vapor guess.

Problem B Excel Checklist — Part 3 of 4

13. Calculate

$$=ZComp(y_i, Temp, Pres, 1) \quad \text{and} \quad =ZComp(x_i, Temp, Pres, 0)$$

for the vapor and liquid compressibility factors.

14. Calculate the fugacity coefficients with

$$=PhiC(y_i, ComponentID, Temp, Pres, ZV)$$

$$=PhiC(x_i, ComponentID, Temp, Pres, ZL)$$

15. Form the K-values from

$$K_i = \frac{\hat{\phi}_i^L}{\hat{\phi}_i^V},$$

then Goal Seek the bubble-point Rachford–Rice residual to zero by changing the bubble-point temperature.

16. Update the vapor composition from

$$y_i = K_i x_i,$$

paste the values back as numbers, and repeat, or use the built-in CHE convergence routine.

17. For the PR dew-point calculation, set

$$\alpha = 1, \quad y_i = z_i,$$

and use the heaviest component as the first liquid guess.

Problem B Excel Checklist — Part 4 of 4

18. Recalculate Z-values, fugacity coefficients, and K-values. Goal Seek the dew-point Rachford–Rice residual to zero by changing the dew-point temperature. Update the liquid composition from

$$x_i = \frac{y_i}{K_i},$$

then repeat, or use the built-in CHE convergence routine. Since the mixture is liquid, use

$$x_i = z_i$$

for the final Peng–Robinson mixture-fugacity calculation. To calculate a pure-component Peng–Robinson fugacity, set the liquid fraction of the chosen component equal to 1 and the other liquid fractions equal to 0, then use

$$f_i^{\text{pure}} = \hat{\phi}_i P.$$

For the mixture, use

$$f_i^{\text{mix}} = \hat{\phi}_i x_i P.$$

Put the final ideal and Peng–Robinson fugacity results into one organized Excel table, and include a separate screenshot of that fugacity table.